

Preliminary Investigation of the Effects of Combining Hanger Reflex and Optical Flow

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Abstract

The Hanger Reflex is a haptic illusion phenomenon in which placing a wire hanger on the head induces involuntary head rotation. This rotation, influenced by skin deformation from pressure, occurs across multiple axes, including yaw, pitch, and roll. In this study, we explored the potential of combining optical flow and the Hanger Reflex in Head-Mounted Displays (HMDs) to manipulate head motion. A preliminary experiment was conducted to investigate the interaction between these two stimuli, and the results suggested that presenting optical flow in a direction different from the yaw-axis rotation induced by the Hanger Reflex could alter the direction of head motion. These findings reveal the complex interaction between tactile and visual feedback and suggest new possibilities for controlling motion presentation in virtual environments, leading to the development of innovative methods for presenting three-axis motion using a single-axis tactile presentation mechanism and addressing issues such as cybersickness.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; **Haptic devices**.

Keywords

Hanger Reflex, Multisensory Interaction, Head-Mounted Display, Perceived Self-Motion

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1 Introduction

Hanger Reflex is a phenomenon where wearing a wire hanger on the head causes involuntary head rotation[1][2]. This phenomenon arises from two-point pressure applied to the head, with the resulting skin deformation contributing to the direction of movement[3][4][5]. Previous study has reported not only yaw-axis rotation but also pitch- and roll-axis movements induced by

lateral skin stretch[6][7]. While there are several studies on navigation and interaction using haptic feedback display based on skin deformation[8][9][10][11][12][13][14], the Hanger Reflex is relatively strong and can directly induce head motion, which make it good candidate for haptic-enabled HMDs[7][15]. However, the strength of the Hanger Reflex along pitch and roll axes are smaller than that around yaw axis, and there are large individual differences.

One way to solve this issue is to combine the Hanger Reflex with other modalities. It has been reported that the magnitude of the motion induced by the Hanger Reflex is influenced by the interpretation of information such as instructions before trials[16]. Furthermore, Sakashita[17] combined the wrist-type Hanger Reflex with electrical muscle stimulation (EMS) for forearm movement, where yaw-axis movement was controlled by the Hanger Reflex and pitch-axis movement by EMS. However, the explicit instruction narrows the application areas, and EMS is not always socially acceptable, especially around head.

Our idea is to combinevection with the Hanger Reflex. Vection is an illusion elicited by flow of visual field (optical flow), and it accompanies involuntary body motion. While its combination with haptic interface has been studied[18][19], they mainly used vibrotactile cues to the sole. Combination ofvection and the Hanger Reflex is expected to offer high convenience, particularly because both can be seamlessly integrated within an HMD.

This paper reports the effect of combiningvection with the Hanger Reflex, especially focusing on the direction of head motion induced by the Hanger Reflex.

2 COMBINED PRESENTATION SYSTEM

2.1 Hanger Reflex Presentation

We developed a Hanger Reflex presentation device inspired by research utilizing air-driven balloons for Hanger Reflex presentation[7] (Figure1). The device consists of a microcontroller (ESP32-DevKitC-32D, Espressif Systems), a pressure sensor (MIS-2503-015G, METRODYNE MICROSYSTEM CORP), an air pump (SC3701PML, SHENZHEN SKOOCOM ELECTRONIC), and a solenoid valve (SC415GF, SHENZHEN SKOOCOM ELECTRONIC), enabling control and communication. The system's refresh rate, encompassing sensing, actuation, and communication, was set to 1.2 kHz. Sensor values obtained via AD converter were stored in a ring buffer of size 9, and the median value was used. The size of the air-driven balloon was 40 mm × 40 mm.

The air-driven balloon achieved a target pressure of 16.510 kPa, with a response time of approximately 30 ms. The target pressure P for the Hanger Reflex was controlled using Equation (1):

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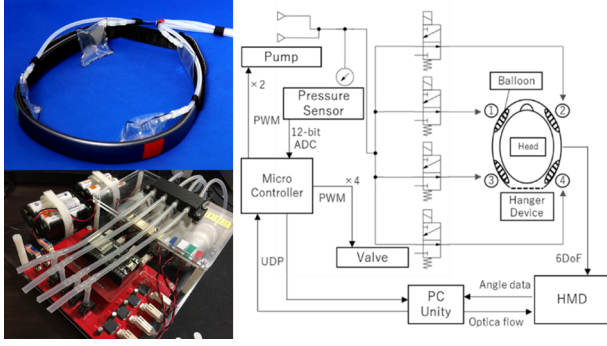


Figure 1: System Overview for Combined Hanger Reflex and Optical Flow Presentation.

$$P = P_{\max} \times \sin\left(2\pi \times \frac{t}{\text{period}} + IP\right) \quad (1)$$

The driving pattern for the Hanger Reflex is shown in Table1. Balloon number in the table corresponds to the number indicated in Figure 1. Pitch is the tilting movement of the head up and down. Roll is the tilting movements of the head side to side. During the pressure increase phase, the balloon induced movements such as Pitch Up, Yaw Right, and Roll Left, while during the decrease phase, it induced Pitch Down, Yaw Left, and Roll Right. Yaw is the rotational movement of the head left and right. For example, by diving balloons 3 and 4, which located at the back of the head, the pitch-up head.

Table 1: Driving Pattern of the Hanger Reflex

Axis	Pitch	Pitch	Yaw	Yaw	Roll	Roll
Direction	Up	Down	Right	Left	Right	Left
Balloon No.	③, ④	①, ②	②, ③	①, ④	②, ④	①, ③

2.2 Optical Flow Presentation

The optical flow stimulus consisted of 2,000 randomly placed spheres, each with a diameter of 0.1 m, within a range of 2.0 m to 5.0 m around the participant's head. The stimuli were created using the Unity game engine (2019.4.11f1, Unity Technologies Inc.) and displayed stereoscopically via an Oculus Rift S HMD (Facebook Technologies, LLC). The HMD operated at a refresh rate of 79–80 FPS.

$$A_{\text{world}} = A_{\max} \times \sin\left(2\pi \times \frac{t}{\text{period}} + IP\right) \quad (2)$$

Equation (2) was used to present optical flow in the Unity environment. A parent object was created to present the optical flow, with the user placed as a child object. In the equation, A_{world} represents the angle of the parent object [degree], where $A_{\max}$ is the maximum rotation angle [degree], t is time [s], period is the cycle duration [s], and IP is the initial phase [rad]. This formula was applied to pitch, yaw, and roll axes. Optical Flow stimuli were presented along axes such as Pitch Down, Yaw Right, and Roll Left

during the air-pressure increase phase, and Pitch Up, Yaw Left, and Roll Right during the air-pressure decrease phase.

2.3 Combined Presentation

Figure 1(right) illustrates the system for combining Hanger Reflex and Optical Flow stimuli. The round-trip communication time, including processing within the Unity-microcontroller system, was measured over 12 trials, yielding an average of 4.583 ms, a minimum of 3 ms, and a maximum of 10 ms, with a standard deviation of 2.290 ms.

3 EXPERIMENT

3.1 Experiment Objective

This preliminary experiment aimed to investigate the interaction between the Hanger Reflex and Optical Flow. The purpose was to examine the parameters, evaluation items, and system configuration for future experiments.

3.2 Experimental Conditions

The experimental conditions are summarized in Table2. For the Hanger Reflex (HR condition), the maximum pressure was set to 16.510 kPa, the period to 10 seconds, and the presentation duration to 30 seconds. The initial phase IP was varied across four conditions: 0 , $\pi/2$, π , and $3\pi/2$. Under the condition where only the Hanger Reflex was tested, it was presented in three directions: pitch axis, yaw axis, and roll axis. Under the condition where optical flow and the Hanger Reflex were combined, the Hanger Reflex was presented only in the yaw axis.

For Optical Flow (OF condition), the maximum rotation angle was set to 90 degrees, the period to 10 seconds, and the presentation duration to 30 seconds. The initial phase IP was also varied across four conditions: 0 , $\pi/2$, π , and $3\pi/2$. Optical Flow was presented independently along the pitch, yaw, and roll axes.

Figure 2 shows an example of combined presentation. In this case, the Hanger Reflex is presented following sine wave ($IP=0$), whereas the optical flow is presented following cosine wave ($IP=3\pi/2$).

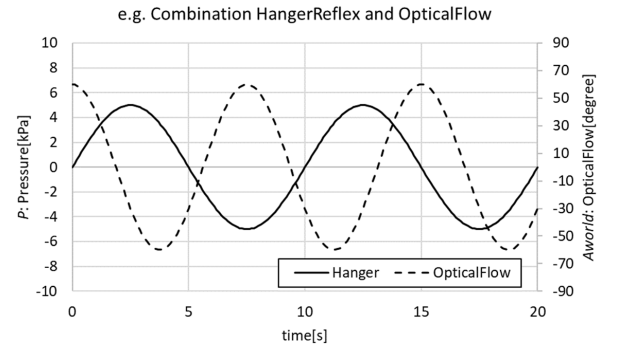


Figure 2: Example of Combined Hanger Reflex and Optical Flow Stimulation Used to Investigate Their Interaction on Head Motion. Hanger Reflex: $P_{\max}=10$, $IP=0$, $\text{period}=10$. OpticalFlow: $A_{\text{world}}=60$, $IP=\pi/2$, $\text{period}=7.5$

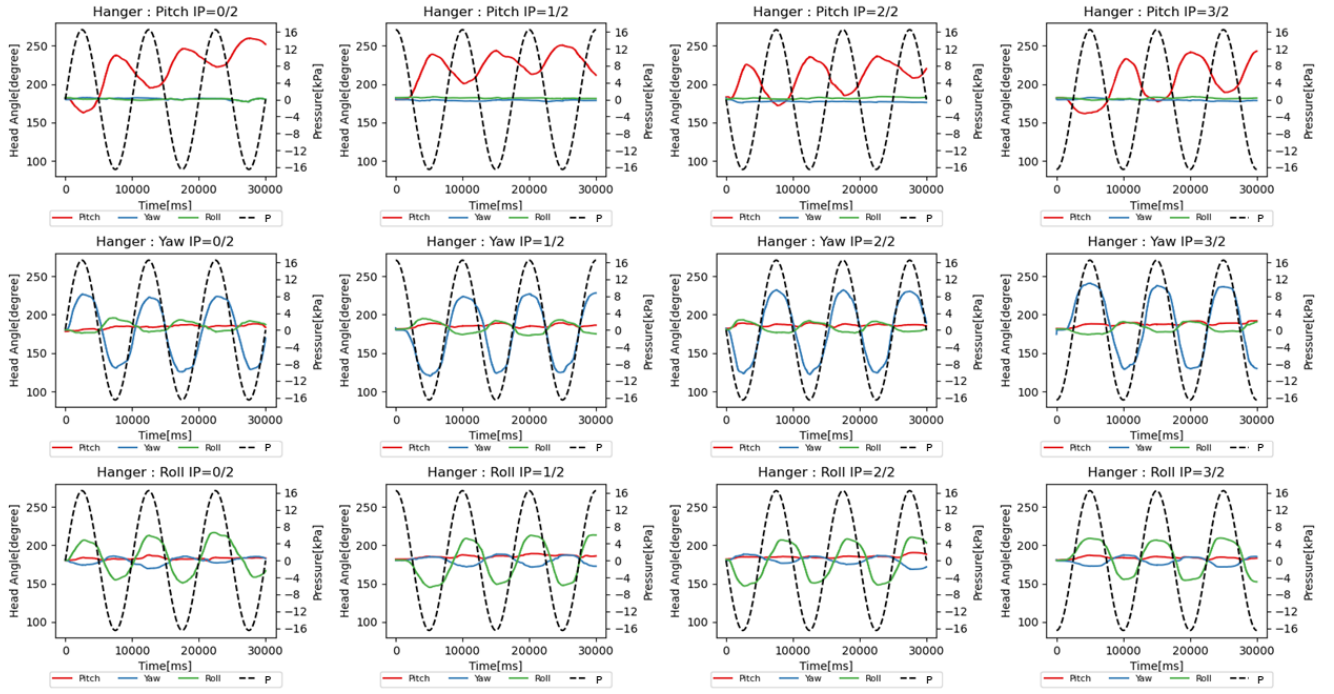


Figure 3: Head Angles Resulting from Varying Hanger Reflex Axis and Initial Phase (IP) Red: Pitch-axis head angle. Blue: Yaw-axis head angle. Green: Roll-axis head angle. Black dashed line: Target pressure (P). Head motion is observed primarily along the axis of Hanger Reflex stimulation.

A total of 72 trials were conducted, combining 48 conditions of HR and OF, and 12 trials each for HR-only and OF-only conditions. The experiment proceeded in the order of HR-only, OF-only, and combined conditions.

Table 2: Presentation Conditions (Excluding Stimulation Axis)

Red: Hanger Reflex only. Blue: Optical Flow only. Yellow: Combined Hanger Reflex and Optical Flow.

		OF condition					
		None	0	$\pi/2$	π	$3\pi/2$	
HR condition	None	N/A	OF(sin)	OF(cos)	OF(-sin)	OF(-cos)	
	0	HR(sin)	HR(sin), OF(sin)	HR(sin), OF(cos)	HR(sin), OF(-sin)	HR(sin), OF(-cos)	
	$\pi/2$	HR(cos)	HR(cos), OF(sin)	HR(cos), OF(cos)	HR(cos), OF(-sin)	HR(cos), OF(-cos)	
	π	HR(-sin)	HR(-sin), OF(sin)	HR(-sin), OF(cos)	HR(-sin), OF(-sin)	HR(-sin), OF(-cos)	
	$3\pi/2$	HR(-cos)	HR(-cos), OF(sin)	HR(-cos), OF(cos)	HR(-cos), OF(-sin)	HR(-cos), OF(-cos)	

3.3 Experimental Procedure

The participant was a 27-year-old male. The Hanger Reflex device and Oculus Rifts HMD were attached to the participant's head, and he was seated in a non-rotating chair. During the trials, the participant was instructed to remain natural and not to resist any induced force or motion. After each trial, the participant returned their head to the forward-facing position. Head angle data was recorded via the HMD. Vection perceived during Optical Flow presentation was responded by pressing a button on a controller (DUALSHOCK 4, Sony Interactive Entertainment Inc.)

3.4 Experimental Results

Figure 3 shows the results when only the Hanger Reflex was presented. The primary vertical axis represents head angle [degree], the secondary vertical axis represents pressure values[kPa], and the horizontal axis represents time[ms]. The first row shows the results for the pitch axis, the second row for the yaw axis, and the third row for the roll axis when the Hanger Reflex was presented. The first column corresponds to the initial phase $IP=0$, the second column to $IP=1\pi/2$, the third column to $IP=\pi$, and the fourth column to $IP=3\pi/2$.

From Figure 3, we observe that motion was induced in the direction of the intended axis when the Hanger Reflex was presented in each axis. When averaging the amplitudes across all initial phase (IP) conditions, the values were 78.388 degrees for the pitch axis, 107.845 degrees for the yaw axis, and 63.481 degrees for the roll axis (Table3).

Table 3: Average Head Angles for Each Presentation Condition

		Condition: HR = HangerReflex, OF = OpticalFlow								
Head Angle [degree]		HR:Pitch	HR:Yaw	HR:Roll	OF:Pitch	OF:Yaw	OF:Roll	HR:Yaw OF:Pitch	HR:Yaw OF:Yaw	HR:Yaw OF:Roll
	Pitch	78.388	9.108	7.506	6.391	2.121	1.799	46.125	7.972	12.801
	Yaw	5.270	107.845	17.085	2.805	4.565	5.032	36.572	100.500	27.834
	Roll	4.095	19.109	63.481	4.179	2.310	9.480	39.124	25.329	75.105

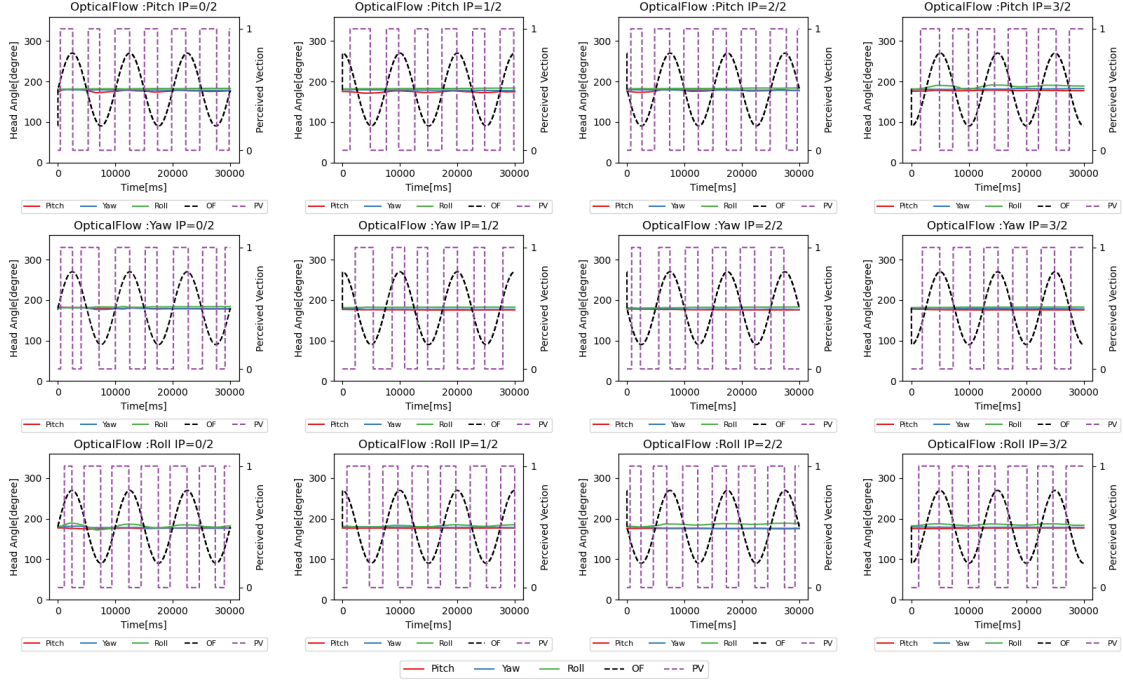


Figure 4: Results of Varying Optical Flow Axis and Initial Phase (IP) Black dashed line: Optical Flow Angle. Purple dashed line: Perceived Vection. Although the induced motion from Optical Flow presentation is minimal, vection perception is confirmed.

Figure 4 shows the results when only optical flow was presented. The primary vertical axis represents head angle [degree], the secondary vertical axis represents the perception of vection (0: no perception, 1: perceived), and the horizontal axis represents time [ms]. The first row shows the results for the pitch axis, the second row for the yaw axis, and the third row for the roll axis when optical flow was presented. The first column corresponds to $IP=0$, the second column to $IP=1\pi/2$, the third column to $IP=\pi$, and the fourth column to $IP=3\pi/2$.

From Figure 4, we confirmed that, under the experimental conditions, head rotation in the pitch, roll, and yaw axes was minimal. However, vection was perceived in alignment with the presentation of optical flow. Averaging the amplitudes across all initial phase IP conditions, the values were 6.391 degrees for the pitch axis, 4.565 degrees for the yaw axis, and 9.480 degrees for the roll axis (Table3).

Figure 5 shows the results when the Hanger Reflex and optical flow were presented together. The vertical axis represents head angle [degree], and the horizontal axis represents time [ms]. The first row shows the results for the pitch axis, the second row for the yaw axis, and the third row for the roll axis when optical flow was presented. The first column corresponds to $IP=0$, the second column to $IP=1\pi/2$, the third column to $IP=\pi$, and the fourth column to $IP=3\pi/2$. In each graph, all initial phase IP conditions of optical flow are combined and displayed in a single plot.

From Figure 5, we observed that when optical flow was presented in the yaw axis, motion induced by the Hanger Reflex occurred without significant influence from the phase parameter of the optical flow. It is noteworthy that, when optical flow was presented in the pitch or roll axes, yaw-axis motion decreased, while motion in

the presented optical flow axis increased. Averaging the amplitudes across all initial phase IP conditions, the values were 46.125 degrees for the pitch axis, 100.500 degrees for the yaw axis, and 75.105 degrees for the roll axis (Table3).

Table3 summarizes the average head angle values for the pitch, yaw, and roll axes under three conditions: Hanger Reflex only (HR), optical flow only (OF), and the combination of Hanger Reflex and optical flow (HR+OF). It was confirmed that head angles in the presented axis were the largest under all conditions. In the HR+OF condition, the head angle in the yaw axis was 100.500 degrees when optical flow was presented in the yaw axis. However, when optical flow was presented in the pitch or roll axes, the head angles in the yaw axis were 46.125 degrees and 75.105 degrees, respectively.

4 DISCUSSION

4.1 Hanger Reflex Presentation

Particularly in the pitch and roll axes, it is observed that the phase of the induced motion lagged the waveform of the target pressure P . This lag can likely be mitigated by advancing the initial phase IP in the Hanger Reflex control equation 1. Additionally, while this experiment controlled the target pressure value P , an alternative approach could involve controlling the pressure based on the user's real-time head angle information by integrating a six-axis sensor into the Hanger Reflex device.

4.2 Optical Flow Presentation

Head movements during the presentation of optical flow were within 10 degrees, but vection in the direction of rotation was still

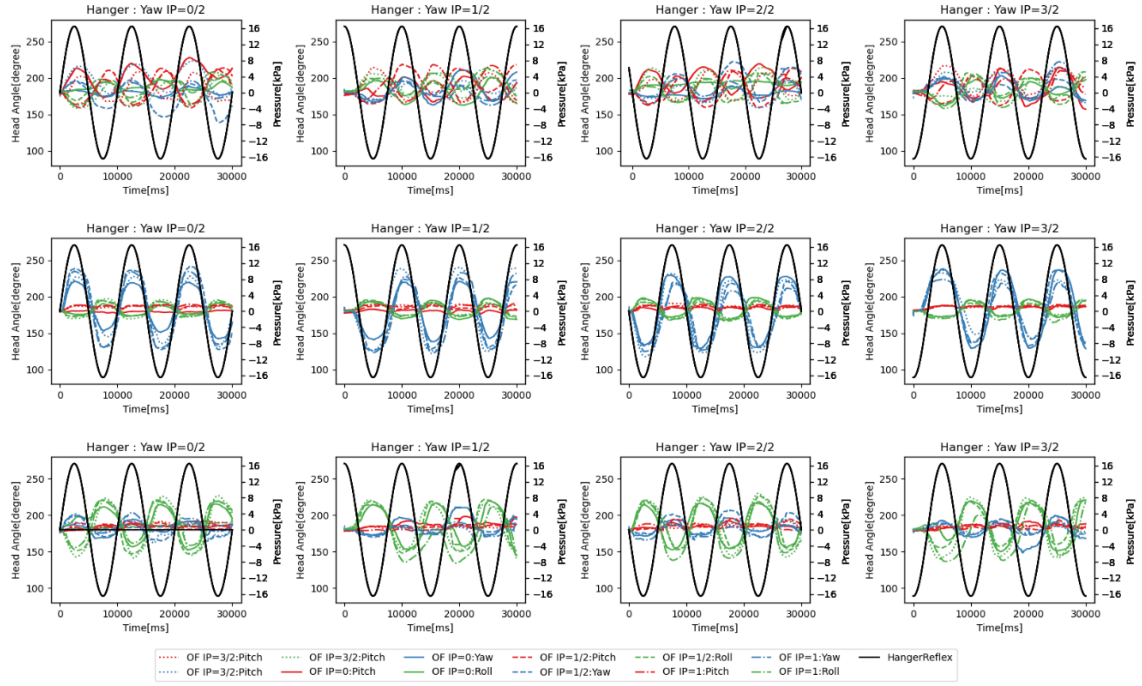


Figure 5: Combination Presentation of Hanger Reflex and Optical Flow

Row 1: Presenting Pitch-axis Optical Flow during Yaw-axis Hanger Reflex induces motion in the Pitch-axis direction.

Row 3: Presenting Roll-axis Optical Flow during Yaw-axis Hanger Reflex induces motion in the Roll-axis direction.

perceived. To modulate the perceived intensity ofvection, it may be possible to controlvection strength by incorporating the effects of various optical flow parameters based on prior research and experimental findings related tovection.

Furthermore, to induce larger rotational movements than observed in this experiment, shortening the period to less than 10 seconds could be a potential approach. However, it is likely that this would increase the probability of VR-induced sickness.

4.3 Combined Presentation

When the Hanger Reflex in the yaw axis was presented in combination with optical flow in the pitch and roll axes, movement was also observed in the direction of the presented optical flow. This suggests the potential to control the movement direction induced by the Hanger Reflex using optical flow. Potential advantages of this approach include: (1) leveraging the high occurrence rate and significant head movement induced by the yaw-axis Hanger Reflex to generate movements in other axes, (2) simplifying the mechanisms for presenting and controlling the Hanger Reflex, and (3) enabling Hanger Reflex control based solely on visual stimuli.

Additionally, if it becomes possible to generate somatosensory output, such as motion, using visual stimuli like optical flow, it may help reduce VR-induced sickness caused by mismatches between visual stimuli and somatosensory feedback.

The timing of the presentation of the Hanger Reflex and optical flow is expected to have a significant influence. While the preliminary experiment used only four conditions for the initial phase

parameter, it is necessary to investigate the effects of more detailed initial phase settings. It is also anticipated that there is a magnitude relationship between the Hanger Reflex and optical flow. For example, if the maximum rotation angle of the optical flow (A_{world}) is reduced, the motion induced in the pitch and roll axes is also expected to decrease.

As shown in the first row of Figure 5, in the condition where optical flow in the pitch axis was presented along with the yaw-axis Hanger Reflex, motion was observed not only in the pitch axis but also in the yaw and roll axes. To reduce unwanted motion in undesired axes, one possible method is to adopt a smaller P_{max} value for the yaw-axis Hanger Reflex than the value used in this experiment.

4.4 Current Limitation and Future Work

This preliminary experiment was conducted with a single participant, necessitating further experiments with an increased number of participants. Additionally, it is worth considering increasing the number of conditions for the initial phase (IP) to examine the effects of the timing of the presentation of the Hanger Reflex and optical flow. By varying parameters such as "amplitude," "frequency," "initial phase," and "waveform" for both the Hanger Reflex and optical flow, future studies aim to investigate their effects on presentation timing, magnitude relationships, and other aspects of interaction.

5 CONCLUSION

This paper reported a preliminary experiment aimed at examining the parameters and evaluation metrics for investigating the effects of combining the Hanger Reflex and optical flow on user perception and head angle. The results of the preliminary experiment confirmed that the motion direction induced by the Hanger Reflex is influenced by the direction of the presented optical flow, suggesting an interaction between the Hanger Reflex and optical flow. Future work will expand the conditions to include amplitude, frequency, and initial phase, and conduct additional experiments to explore factors such as magnitude relationships, thresholds, and contributing parameters in the interaction between the Hanger Reflex and optical flow.

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